

Reseto

Day 5 solution walkthrough

Competitive Programming Bootcamp

Statement excerpts:

- ▶ *“ K -th integer to be crossed out”*
- ▶ *“Cross out P and all its multiples”*

Programming interpretation:

- ▶ The problem asks for the simulation order of the sieve, not just primes.
- ▶ Count a number only the first time it is crossed out.
- ▶ Stop immediately when the K -th crossing occurs.

Textbook connection: CP4 5.3.1 Prime Numbers

Core technique: Sieve simulation

- ▶ Simulate crossing numbers out of the Sieve of Eratosthenes.
- ▶ Count each number the first time it is crossed out.
- ▶ Print the K-th crossed-out number.

- ▶ This is not just prime generation; the crossing order matters.
- ▶ The prime itself is counted when crossed.
- ▶ Already crossed multiples must not be counted again.

- ▶ Create `crossed[0..N]` initialized false.
- ▶ For p from 2 to N , skip p if it is already crossed.
- ▶ For each multiple of p starting at p , skip if already crossed.
- ▶ Mark the multiple, increment the crossing count, and stop when `count == K`.

- ▶ Processing p in increasing order matches the sieve rule.
- ▶ Starting at p , not p^*p , is required because the problem counts p itself.
- ▶ The first time a number is crossed determines its count position.

- ▶ The outer loop always chooses the smallest not-yet-crossed number.
- ▶ The inner loop crosses exactly its still-available multiples in increasing order.
- ▶ Stopping at the K-th crossing returns the number requested by the problem statement.

Complexity: $O(N \log \log N)$ sieve-style time and $O(N)$ memory.

- ▶ This differs from an optimized prime-only sieve because output depends on crossing order.
- ▶ Do not count a number twice.

Source lines: `reseto.py:36-49`

```
import sys

def main():
    input = sys.stdin.buffer.readline

    n, k = map(int, input().split())

    crossed = [False] * (n + 1)

    count = 0
```

What this chunk does

- ▶ `n` is the sieve limit and `k` is the requested crossing count.
- ▶ `crossed` marks whether each integer has already been removed.
- ▶ `count` tracks the order of removals.

Source lines: `reseto.py:51-68`

```
for p in range(2, n + 1):  
    if crossed[p]:  
        continue  
  
    for multiple in range(p, n + 1, p):  
        if crossed[multiple]:  
            continue
```

What this chunk does

- ▶ The outer loop finds the next uncrossed number p .
- ▶ Already crossed p values cannot be the next prime in the simulation.
- ▶ The inner loop visits p itself and then every multiple of p .

Source lines: `reseto.py:70-80`

```
        crossed[multiple] = True
        count += 1

    if count == k:
        print(multiple)
        return

main()
```

What this chunk does

- ▶ Only newly crossed numbers increment the counter.
- ▶ The moment count equals k, that multiple is the answer.
- ▶ Returning immediately avoids extra sieve work.

Solution file: `practice_contests/day_5/reseto.py`

Problem: <https://open.kattis.com/problems/reseto>

Use this deck with the source code open beside it. The slides explain the reasoning; the code shows the exact input handling and output formatting.